

LA HW

Included exercises: 15, 18-19, 21, 23, 25, 43, 47-48, 55-56, 59, 61, 64

Exercises 13-20

In Exercises 13–20, a symmetric matrix A is given. Find an orthonormal basis of eigenvectors and their corresponding eigenvalues. Use this information to obtain a spectral decomposition of each matrix.

Original Exercise 15

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

Original Exercise 18

$$\begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{bmatrix}$$

Original Exercise 19

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Exercises 21-40

In Exercises 21–40, determine whether the statements are true or false.

Original Exercise 21

Every symmetric matrix is diagonalizable.

Original Exercise 23

If A is an $n \times n$ matrix and there exists an orthonormal basis for $\mathcal{R}^n$ consisting of eigenvectors of A , then A is

Original Exercise 25

Distinct eigenvectors of a symmetric matrix are orthogo-

Original Exercise 43

Exercise 19.

Exercises 47-54

In Exercises 47–54, let A be an $n \times n$ symmetric matrix with spectral decomposition $A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_n P_n$. Assume that $\mu_1, \mu_2, \dots, \mu_k$ are all the distinct eigenvalues of A and that Q_i denotes the sum of all the P_i 's that are associated

Original Exercise 47

u_j .

Prove $A = \mu_1 Q_1 + \mu_2 Q_2 + \cdots + \mu_k Q_k$.

Original Exercise 48

Prove $Q_j Q_j = Q_j$ for all j , and $Q_i Q_j = O$ if $j \neq i$.

Exercises 55-57

Exercises 55–57 use the definition: For an $n \times n$ matrix B and a polynomial $g(t) = a_n t^n + a_{n-1} t^{n-1} + \cdots + a_1 t + a_0$, define the matrix $g(B)$ by

Original Exercise 55

Use Exercises 47 and 48 to show that for any polynomial g ,

$$g(A) = g(\mu_1)Q_1 + g(\mu_2)Q_2 + \cdots + g(\mu_k)Q_k,$$

Original Exercise 56

where A , μ_i , and Q_i are as in Exercises 47–54.

Use Exercise 55 to prove a special case of the Cay-

Exercises 59-73

Prove that P is the orthogonal projection matrix for the subspace $\text{Span}\{\mathbf{u}\}$.

Original Exercise 59

ion $\mathbf{v}^T A \mathbf{v} = \mathbf{v} \cdot A \mathbf{v}$ is often useful in solving these exercises.

Suppose that A is a symmetric matrix. Prove that A is positive definite if and only if all of its eigenvalues are

Original Exercise 61

matrices analogous to that in Exercise 59.

Suppose that A is invertible and positive definite. Prove

Original Exercise 64

Suppose that A and B are positive definite $n \times n$ matrices.